

Shortest path

Longest path

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Optimization: principles and algorithms

Longest path

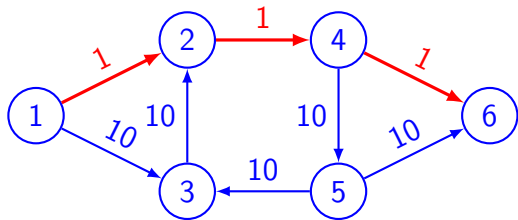
In optimization, we can change a minimization problem into a maximization problem.

We analyze the implications for the longest path problem.

Definition

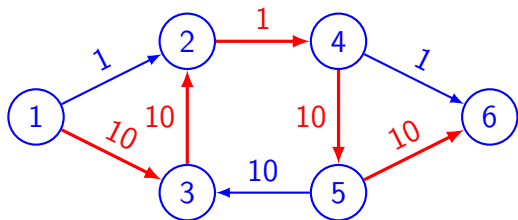
Shortest path problem

Find a simple forward path between o and d with the **minimal** cost.



Longest path problem

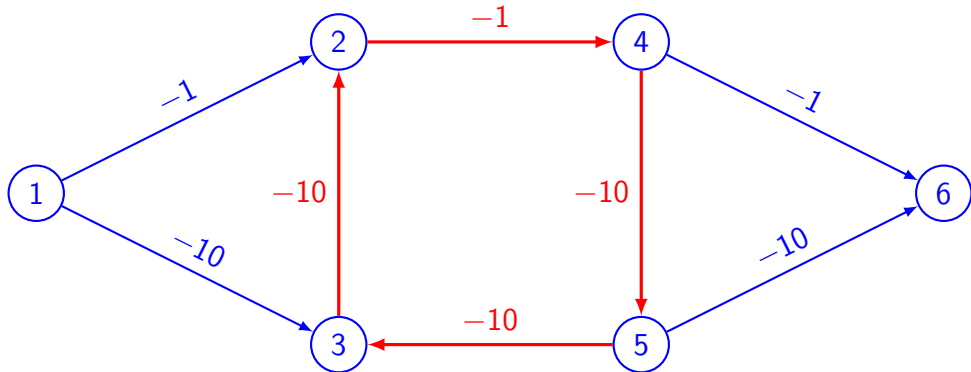
Find a simple forward path between o and d with the **maximal** cost.



Transshipment problem

$$\max_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij} \iff \min_{x \in \mathbb{R}^n} - \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij} = \sum_{(i,j) \in \mathcal{A}} (-c_{ij}) x_{ij}.$$

Risk of negative cycle



Summary

The longest path problem is defined in a similar way as the shortest path problem. As a maximization problem can be transformed in a minimization problem by changing the sign of the objective function, it can also be solved with the same algorithms.

However, the shortest path algorithm assumes that there is no cycle of negative cost.

For many applications, the longest path problem cannot rely on these algorithms because of the presence of a negative cost cycle once the signs have been changed.

In the next video, we see an application where there is no such cycle.